

Test Case Evaluation and Execution Logs for ScreenPilot System

Test Case 01: Writing on Notepad (Simple Task)

User Prompt:

Type "*Agent*" in Notepad

Test Results:

- Steps Total: 1
- Completed: 1
- Failed: 0
- Duration: 12 seconds

Step Execution:

Step 1: Type "*Agent*" into the Notepad text area

- Target: Notepad text area
- Attempts: 1
- HID Commands:
 - `move_mouse(x, y)`
 - `click()`
 - `type_text("Agent")`
- Result: Successfully executed

Execution Summary:

The system successfully detected the Notepad editor region and generated valid HID commands. The cursor was positioned correctly, and the text "*Agent*" was entered without errors. Post-execution validation confirmed successful task completion.

Success Factors:

- Clear visibility of the Notepad window
- Accurate detection of the text input region
- Valid and complete HID command generation
- Effective closed-loop validation

JSON Response:

```
{
  "run_id": "d4d47075-2ca8-4de5-babe-c7783e4fec00",
  "user_task": "type \"Agent\" on the notepad",
```

```
"started_at": "2026-04-29T16:25:04.284619+00:00",
"finished_at": "2026-04-29T16:25:16.284619+00:00",
"vision_used": true,

"perception": {
  "status": "completed",
  "session_id": "shot_1777479912_781bf2",
  "vision_data": {
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "notepad_editor",
        "type": "text_area",
        "label": "Notepad Editor",
        "description": "Main text editing area",
        "state": "visible",
        "dx": 500,
        "dy": 400,
        "confidence": 0.95,
        "source": "gemini_vlm"
      }
    ],
    "element_count": 1
  }
},

"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Type 'Agent' into the notepad text area.",
      "target": "Notepad text area",
      "expected_result": "The word 'Agent' is visible in the notepad
document.",
      "status": "completed"
    }
  ],
  "estimated_complexity": "simple"
},

"step_plans": [
```

```

{
  "step_id": 1,
  "action": "Type 'Agent' into the notepad text area.",
  "target": "Notepad text area",
  "hid_commands": [
    { "action": "move_mouse", "x": 500, "y": 400 },
    { "action": "click" },
    { "action": "type_text", "text": "Agent" }
  ],
  "reasoning": "Detected text area and generated valid HID commands."
},
],
"step_results": [
  {
    "step_id": 1,
    "status": "success",
    "executed_commands": 3,
    "verification": "Text 'Agent' successfully entered into Notepad."
  }
],
"final_report": {
  "success": true,
  "summary": "The task was successfully completed.",
  "message": "The system detected the Notepad editor, positioned the cursor, and typed 'Agent' successfully.",
  "steps_completed": 1,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
}

```

Test Case 02: Screen Analysis (Medium Task)

User Prompt:

Test the current screen and explain important elements

Test Results:

- Steps Total: 1
- Completed: 1
- Failed: 0

- Duration: 5 minutes 36 seconds

Step Execution:

Step 1: Analyze and describe the current screen

- Target: Entire screen
- Attempts: 1
- Execution: Success
- Evaluation: Completed

Execution Summary:

The agent successfully analyzed the screen and identified key elements. It interacted with the interface by clicking the address bar to confirm interactivity and extracted contextual information, including the active URL (*16.16.24.81:3000/analytics*).

Observations:

- Demonstrates strong perception + reasoning capability
- Slight latency due to vision and LLM processing stages

JSON Response:

```
{
  "run_id": "c3d74960-0d60-43a0-8b50-0178b3905308",
  "user_task": "test the current screen and explain important things about that screen",
  "started_at": "2026-04-29T19:11:29.400364+00:00",
  "finished_at": "2026-04-29T19:17:05.979272+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777489894_25f120\\processing\\proc_1777489894_e33d95\\preprocessed_frames\\frame_1777489894_bef67e.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      115,
      65,
      1805,
      1015
    ]
  }
}
```

```
"screen_size": {
  "width": 1690,
  "height": 950
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "browser_address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing the URL
16.16.24.81:3000/analytics",
    "state": "enabled",
    "dx": 214,
    "dy": 30,
    "frame_dx": 329,
    "frame_dy": 95,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      201,
      78,
      458,
      111
    ]
  },
  {
    "id": "nexmart_logo",
    "type": "icon",
    "label": "NexMart",
    "description": "NexMart logo and home link in the navigation bar",
    "state": "enabled",
    "dx": 221,
    "dy": 90,
    "frame_dx": 336,
    "frame_dy": 155,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      280,
      138,
      392,
      171
    ]
  }
],
},
```

```
{
  "id": "nav_analytics",
  "type": "link",
  "label": "Analytics",
  "description": "Navigation link to the Analytics page",
  "state": "active",
  "dx": 472,
  "dy": 92,
  "frame_dx": 587,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    551,
    144,
    623,
    171
  ]
},
{
  "id": "nav_products",
  "type": "link",
  "label": "Products",
  "description": "Navigation link to the Products page",
  "state": "enabled",
  "dx": 571,
  "dy": 92,
  "frame_dx": 686,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    650,
    144,
    722,
    171
  ]
},
{
  "id": "nav_orders",
  "type": "link",
  "label": "Orders",
  "description": "Navigation link to the Orders page",
  "state": "enabled",
  "dx": 657,
```

```
"dy": 92,
"frame_dx": 772,
"frame_dy": 157,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  742,
  144,
  802,
  171
]
},
{
  "id": "nav_payments",
  "type": "link",
  "label": "Payments",
  "description": "Navigation link to the Payments page",
  "state": "enabled",
  "dx": 746,
  "dy": 92,
  "frame_dx": 861,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    821,
    144,
    901,
    171
  ]
}
],
"element_count": 6
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "test the current screen and explain important things about
that screen",
      "target": "screen",
      "expected_result": "Task accomplished",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    }
  ]
}
```

```
    }
  ],
  "notes": "Fallback single-step plan (LLM error: Response missing valid 'steps' list. Got: NoneType)",
  "estimated_complexity": "unknown"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "test the current screen and explain important things about that screen",
    "target": "screen",
    "expected_result": "Task accomplished",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 299,
        "dy": 24
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ]
  },
  {
    "reasoning": "Clicking the address bar to verify interactivity and focus as part of testing the screen elements."
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "test the current screen and explain important things about that screen",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "done",
    "confidence": 1.0,
    "reason": "The screen has been successfully analyzed, identifying the browser address bar and the current URL as the analytics page (16.16.24.81:3000/analytics).",
    "final_status": "executing",
    "timestamp": "2026-04-29T19:16:19.405915+00:00"
  }
],
"final_report": {
```



```
    "success": true,
    "summary": "Successfully analyzed the screen and identified key navigation elements for NexMart.",
    "message": "The task to test and explain the current screen has been completed successfully.\n\nAnalysis of the screen reveals that this is the NexMart interface. Key elements identified include the 'NexMart' logo text and primary navigation links for 'Analytics' and 'Products'.\n\nStep results:\n- Step 1: test the current screen and explain important things about that screen (Completed)",
    "steps_completed": 1,
    "steps_failed": 0,
    "issues": [],
    "recommendations": []
  }
}
```

Test Case 03: Login UI Test (Complex Task)

User Prompt:

Login using username = *nuwani* and password = *Nuwani@345*

Test Results:

- Steps Total: 2
- Completed: 0
- Failed: 2
- Duration: 13 minutes 39 seconds

Step Execution:

Step 1: Enter credentials and click Sign In

- Target: NexMart login form
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)

Issue:

The system executed input actions but failed to confirm successful login due to invalid evaluation output (malformed JSON).

Step 2: Verify dashboard loaded

- Target: NexMart dashboard
- Execution: Failed
- Issue: No `hid_commands` generated

Execution Summary:

Although the system performed initial interactions (typing and clicking), it failed to complete the task due to issues in the evaluation and planning stages. The workflow stalled due to missing or malformed LLM outputs.

Issues Identified:

- Malformed JSON response from LLM during evaluation
- Missing `hid_commands` in step planning
- No transition from login screen to dashboard
- Possible UI ambiguity or incorrect element targeting

Recommendations for Improvement:

- Enforce strict JSON output formatting in LLM responses
- Implement retry and fallback mechanisms for missing commands
- Improve vision prompts for better UI element detection
- Strengthen evaluation loop to handle incomplete responses
- Add validation for dynamic UI behaviors (e.g., errors, CAPTCHA)

JSON Response:

-

full_cycle.json

```
{
  "run_id": "fb5c82a7-0146-427f-9d46-304f4de95361",
  "user_task": "login to the system username = nuwani and the password = Nuwani@345",
  "started_at": "2026-04-21T11:08:00.575402+00:00",
  "finished_at": "2026-04-21T11:21:39.710276+00:00",
```

```
"vision_used": true,
"perception": {
  "status": "completed",
  "session_id": "shot_1776769686_4fe97d",
  "source_frame": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\ra
w_frames\\frame_1776769686_cadbce.jpg",
  "processed_at": 1776769794.6070378,
  "final_json_path": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\fi
nal_elements\\frame_1776769686_cadbce.json",
  "vision_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\pr
ocessing\\proc_1776769686_5165f7\\preprocessed_frames\\frame_1776769686_cadbce.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "0",
        "type": "tab",
        "label": "NexMart",
        "description": "Browser tab for NexMart",
        "state": "active",
        "dx": 150,
        "dy": 45,
        "confidence": 0.9,
        "source": "gemini_vlm"
      },
      {
        "id": "1",
        "type": "tab",
        "label": "Google Password Manager",
        "description": "Browser tab for Google Password Manager",
        "state": "inactive",
        "dx": 400,
        "dy": 45,
        "confidence": 0.9,
        "source": "gemini_vlm"
      },
      {
```

```
"id": "2",
"type": "tab",
"label": "frontend",
"description": "Browser tab for frontend",
"state": "inactive",
"dx": 650,
"dy": 45,
"confidence": 0.9,
"source": "gemini_vlm"
},
{
  "id": "3",
  "type": "tab",
  "label": "Ask Gemini",
  "description": "Browser tab for Ask Gemini",
  "state": "inactive",
  "dx": 1500,
  "dy": 45,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "4",
  "type": "input",
  "label": "Address Bar",
  "description": "Browser address bar showing URL",
  "state": "visible",
  "dx": 400,
  "dy": 85,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "5",
  "type": "icon",
  "label": "NexMart Logo",
  "description": "Logo for NexMart application",
  "state": "visible",
  "dx": 840,
  "dy": 190,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "6",
```

```
    "type": "label",
    "label": "NexMart",
    "description": "Application name text",
    "state": "visible",
    "dx": 840,
    "dy": 230,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "7",
    "type": "label",
    "label": "Welcome back",
    "description": "Login card header text",
    "state": "visible",
    "dx": 840,
    "dy": 330,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "8",
    "type": "input",
    "label": "Username Input",
    "description": "Input field for username",
    "state": "empty",
    "dx": 840,
    "dy": 420,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "9",
    "type": "input",
    "label": "Password Input",
    "description": "Input field for password",
    "state": "empty",
    "dx": 840,
    "dy": 490,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "10",
    "type": "button",
```

```
    "label": "Sign In",
    "description": "Button to submit login credentials",
    "state": "visible",
    "dx": 840,
    "dy": 550,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "11",
    "type": "link",
    "label": "Create one",
    "description": "Link to create a new account",
    "state": "visible",
    "dx": 840,
    "dy": 590,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "12",
    "type": "button",
    "label": "Start Button",
    "description": "Windows Start menu button",
    "state": "visible",
    "dx": 470,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "13",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows taskbar search bar",
    "state": "visible",
    "dx": 575,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "14",
    "type": "icon",
    "label": "Chrome Icon",
```

```
    "description": "Google Chrome icon on taskbar",
    "state": "visible",
    "dx": 1020,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "15",
    "type": "icon",
    "label": "VS Code Icon",
    "description": "Visual Studio Code icon on taskbar",
    "state": "visible",
    "dx": 1105,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "16",
    "type": "label",
    "label": "Clock",
    "description": "System clock and date on taskbar",
    "state": "visible",
    "dx": 1565,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "17",
    "type": "icon",
    "label": "Weather Widget",
    "description": "Weather widget on taskbar",
    "state": "visible",
    "dx": 120,
    "dy": 825,
    "confidence": 0.9,
    "source": "gemini_vlm"
  }
],
"element_count": 18
}
},
"todo": {
```

```
"steps": [
  {
    "id": 1,
    "action": "Click the 'Username Input' field, type 'nuwani', click the 'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",
    "target": "NexMart login form",
    "expected_result": "The login form is submitted and the application redirects to the dashboard.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  },
  {
    "id": 2,
    "action": "Verify that the dashboard or home page has loaded to confirm successful authentication.",
    "target": "NexMart dashboard",
    "expected_result": "The user is successfully logged in and viewing the main application interface.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
"notes": "The screen shows the NexMart login page. The user has provided the necessary credentials in the task description, so no further user input is required to complete the login process.",
"estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click the 'Username Input' field, type 'nuwani', click the 'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",
    "target": "NexMart login form",
    "expected_result": "The login form is submitted and the application redirects to the dashboard.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 848,
        "dy": 586
      },
      {
        "cmd": "mouse_click",
```



```

        "button": "left"
    },
    {
        "cmd": "type_text",
        "text": "nuwani"
    },
    {
        "cmd": "mouse_move",
        "dx": 839,
        "dy": 692
    },
    {
        "cmd": "mouse_click",
        "button": "left"
    },
    {
        "cmd": "type_text",
        "text": "Nuwani@345"
    },
    {
        "cmd": "mouse_move",
        "dx": 875,
        "dy": 838
    },
    {
        "cmd": "mouse_click",
        "button": "left"
    }
],
    "reasoning": "Focusing the username field to type 'nuwani', then focusing
the password field to type 'Nuwani@345', and finally clicking the sign-in button
to submit the form."
},
{
    "step_id": 2,
    "action": "Verify that the dashboard or home page has loaded to confirm
successful authentication.",
    "target": "NexMart dashboard",
    "expected_result": "The user is successfully logged in and viewing the main
application interface.",
    "hid_commands": [],
    "reasoning": "Command generation failed: Response missing 'hid_commands'"
}
],
    "step_results": [

```

```

{
  "step_id": 1,
  "action": "Click the 'Username Input' field, type 'nuwani', click the
'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",
  "attempt": 3,
  "exec_status": "success",
  "eval_status": "retry",
  "confidence": 0.0,
  "reason": "Evaluation error: No valid JSON found in LLM response:
'{\\n  \\\"status\\\": \\\"retry\\\",',",
  "final_status": "executing",
  "timestamp": "2026-04-21T11:17:33.798945+00:00"
}
],
"final_report": {
  "success": false,
  "summary": "The login attempt failed, and the application remains on the
login screen.",
  "message": "The task to log in to the NexMart system was unsuccessful.
Although the steps were initiated, the system failed to process the credentials
and navigate to the dashboard. The final screen state confirms that the 'Welcome
back' login page is still active.\\n\\nStep results:\\n- Step 1: Enter credentials
and click 'Sign In' \\u2014 FAILED\\n- Step 2: Verify dashboard loading \\u2014
FAILED",
  "steps_completed": 0,
  "steps_failed": 2,
  "issues": [
    "The automation failed to successfully submit the login form.",
    "The screen state did not change from the login page to the dashboard."
  ],
  "recommendations": [
    "Verify that the username 'nuwani' and password 'Nuwani@345' are correct.",
    "Check for any hidden error messages or validation alerts on the page.",
    "Ensure the 'Sign In' button is clickable and the application is
responsive."
  ]
}
}

```